

Decode Labs

Cyber Security Internship

PROJECT 4

Vulnerability Assessment Report

Windows System Security Audit

Submitted By

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Batch

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1. Introduction

Cybersecurity is the practice of protecting computer systems, networks, applications, and data from unauthorized access, misuse, damage, or disruption. Modern computer systems face various security threats, including malware infections, unauthorized account access, data theft, and exploitation of software vulnerabilities.

A vulnerability assessment is a systematic process used to identify security weaknesses within a system and evaluate the associated risks. It helps organizations and individuals understand their current security posture and implement appropriate security controls.

This project focuses on performing a basic security audit of a Windows-based computer system. The assessment examines authentication security, software updates, endpoint protection, user security practices, and system configuration settings. The purpose is to identify weaknesses and provide recommendations to improve the overall security of the system.

2. Objectives

The objectives of this security assessment are:

- To evaluate the security configuration of a Windows operating system.
- To identify weaknesses in authentication and access control.
- To verify that security updates and software patches are installed.
- To check antivirus and firewall protection.
- To examine user security practices and system settings.
- To analyze identified risks and provide recommendations for mitigation.

3. Scope of Assessment

The assessment was performed on a Windows computer system. The following security areas were evaluated:

- User authentication and account protection
- Multi-Factor Authentication (MFA)
- Windows Update and patch management
- Installed applications
- Antivirus protection
- Firewall configuration
- Disk encryption
- Browser security

- Startup applications
- User Account Control (UAC)

4. Methodology

The vulnerability assessment was conducted using manual inspection of Windows security settings and built-in system tools. The following Windows features were examined:

- Windows Settings
- Windows Security
- Windows Update
- Installed Apps
- Task Manager
- User Account Control settings

Screenshots were collected as evidence during the assessment process.

5. Security Assessment Results

5.1 Identity and Authentication Security

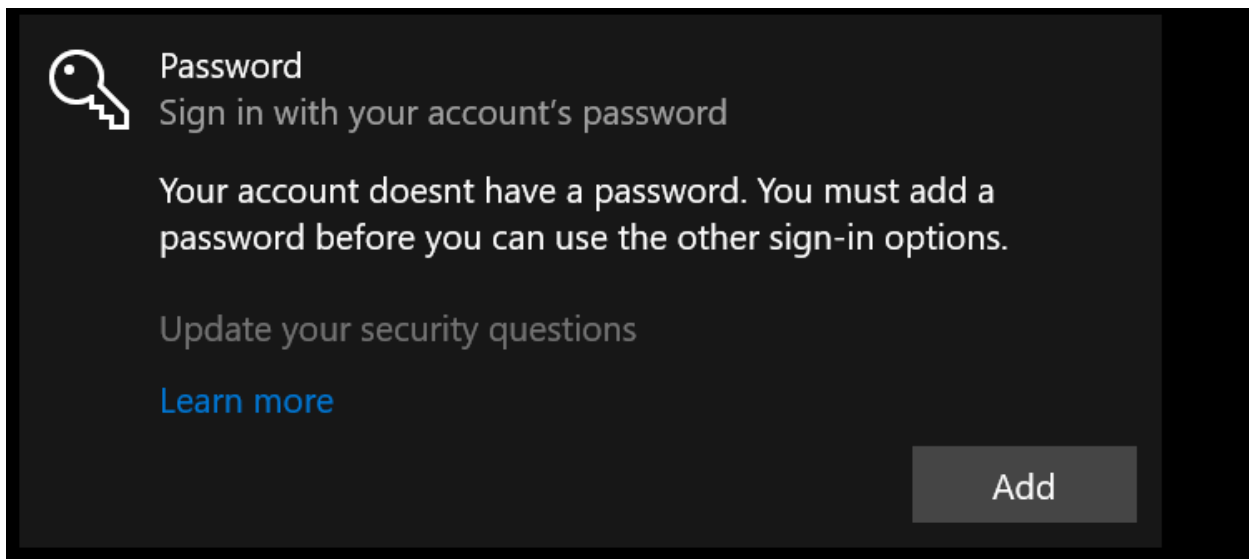
Authentication is the process of verifying the identity of a user before allowing access to a system. Strong authentication mechanisms prevent unauthorized users from accessing sensitive information.

Assessment

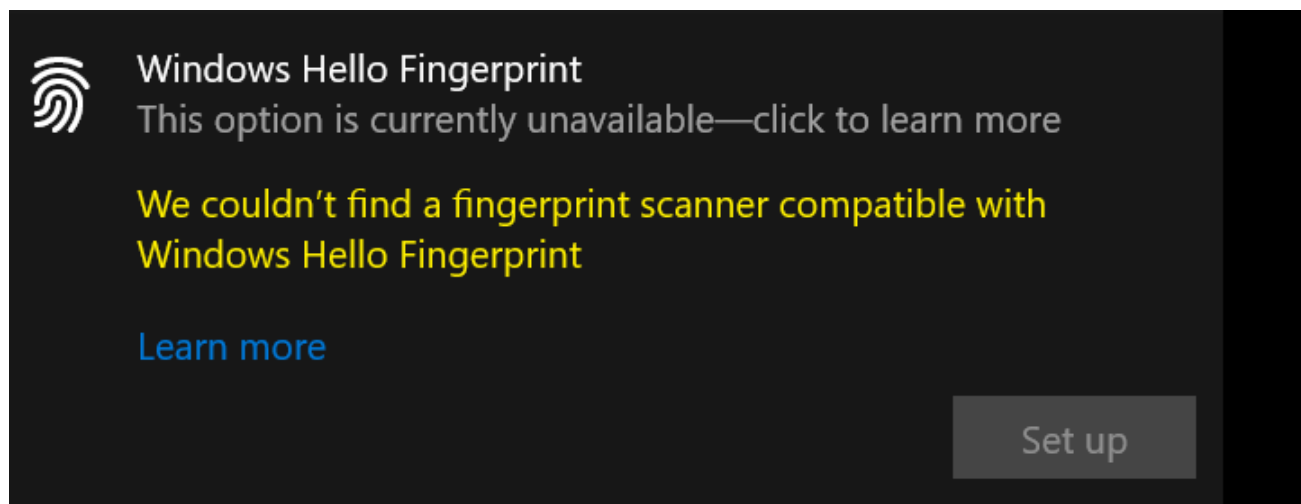
Security Control	Status
Windows Password	Not Configured
Windows Hello PIN	Not Configured
Fingerprint Authentication	Unavailable
Face Recognition	Unavailable
Google MFA	Disabled
Instagram MFA	Disabled

Evidence

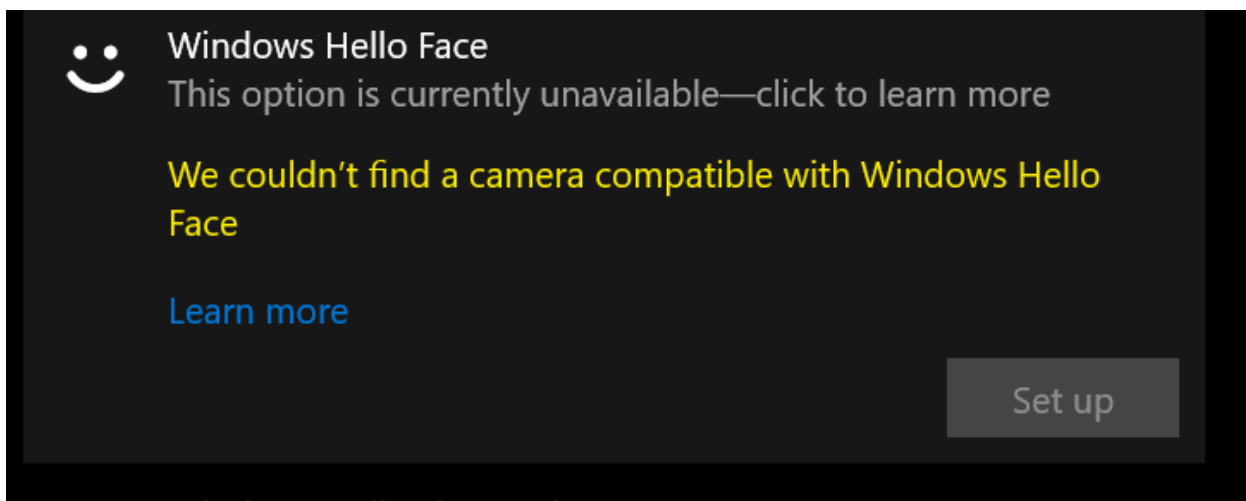
1. Windows Sign-in Options:



1.1 Password

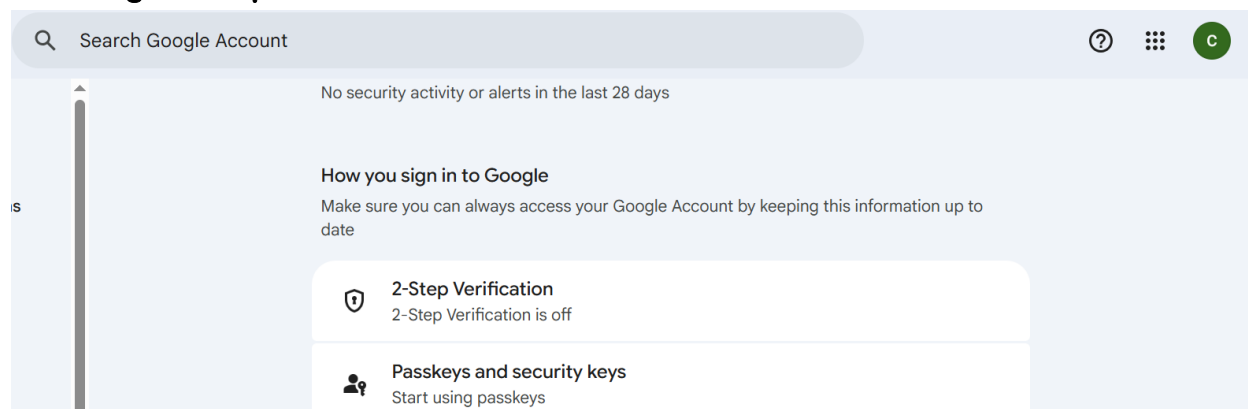


1.2 Fingerprint

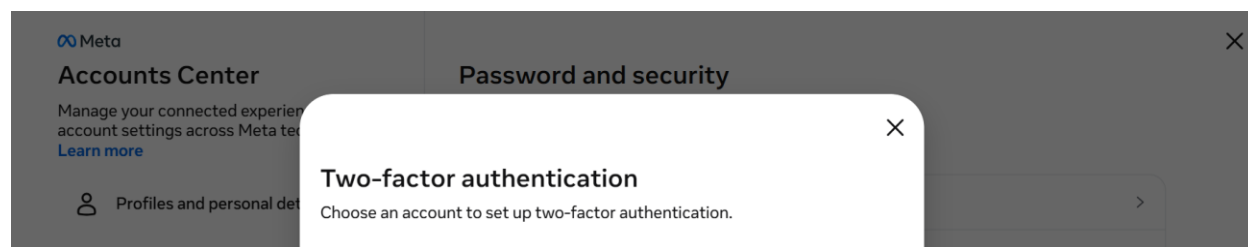


1.3 Face Recognition

2. Google 2-Step Verification Status



3. Instagram Two-Factor Authentication Status



Findings

Finding 1: No Windows Password Configured

Risk Level: High

The Windows account does not have a password configured. Without password protection, anyone with physical access to the computer may access user files and settings.

Recommendation: A strong password should be configured. Windows Hello PIN can also be enabled after setting a password.

Finding 2: Multi-Factor Authentication Disabled

Risk Level: Medium

MFA adds an additional security layer by requiring another verification method besides a password. Both Google and Instagram accounts were found without MFA protection.

Recommendation: Enable MFA on all important online accounts to reduce the risk of unauthorized access.

5.2 Software and Patch Management

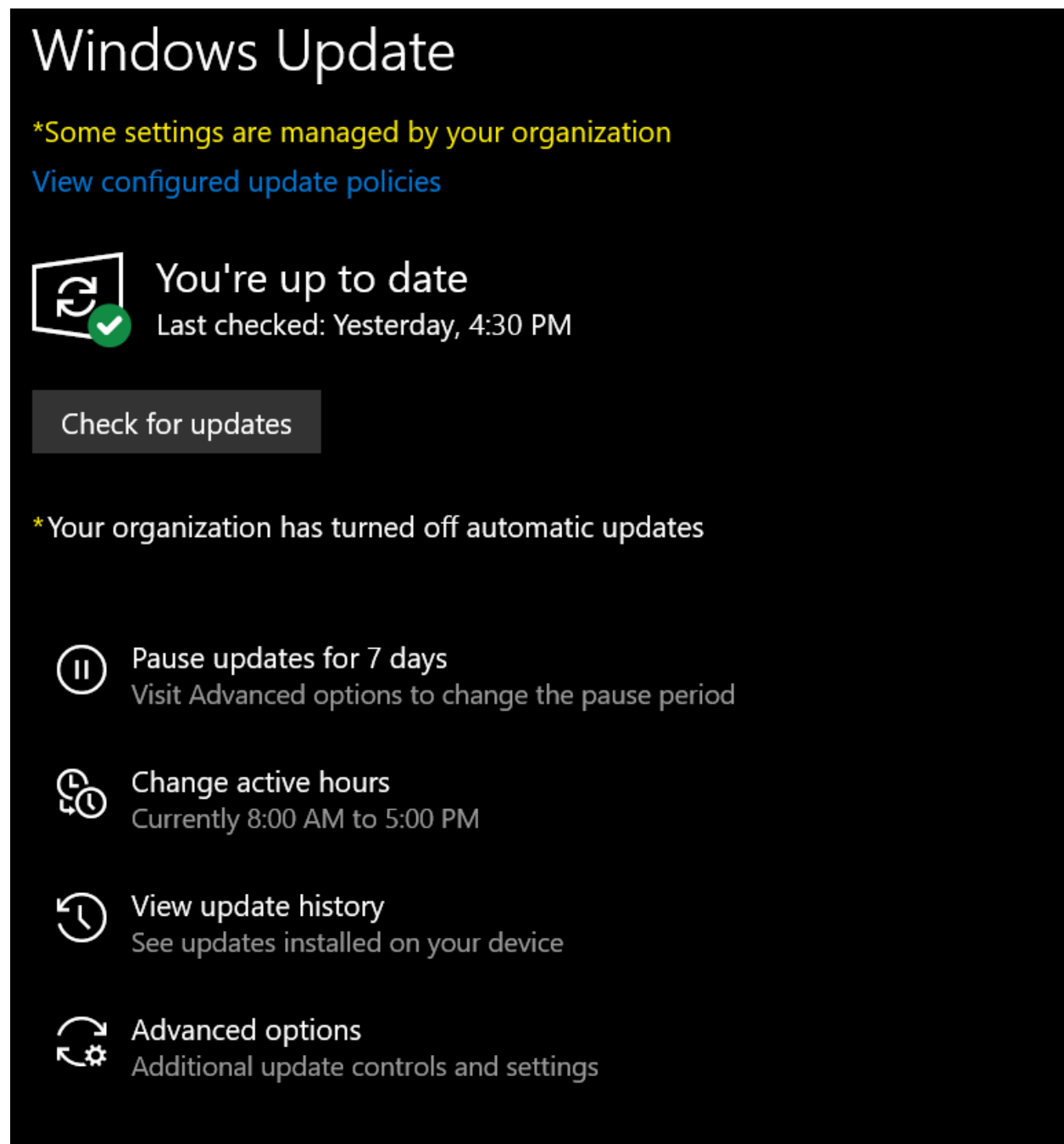
Software patch management involves keeping operating systems and applications updated to protect against known vulnerabilities. Outdated software may contain security weaknesses that attackers can exploit.

Assessment

Security Control	Status
Windows Update	Up to Date
Last Update Check	Yesterday
Update History	No recent entries displayed
Driver Updates	Available section present
Installed Applications	Multiple applications installed
Outdated Software	Not Identified

Evidence

4. Windows Update Page and Update History



5. Installed Applications List

Apps & features

	Visual Studio Code Microsoft Corporation	550 MB 7/7/2026
	Google Chrome	474 MB 7/10/2026
	HEIF Image Extension Microsoft Corporation	16.0 KB 6/20/2026
	Intel® Graphics Command Center INTEL CORP	8.00 KB 11/22/2025
	IntelliJ IDEA 2025.3.2	1/31/2026
	Internet Download Manager	6/5/2023
	Java(TM) SE Development Kit 25.0.2 (64-bit)	364 MB 2/1/2026
	K-Lite Mega Codec Pack 17.4.1	188 MB 6/5/2023
	Mail and Calendar Microsoft Corporation	4.97 MB 11/22/2025
	Maps Microsoft Corporation	16.0 KB 11/22/2025

6. Findings

Finding: Patch Management Status

Risk Level: Low

The operating system reports that it is up to date. No obviously outdated applications were identified during the review.

Recommendation: Continue enabling automatic updates and regularly review installed software.

5.3 Endpoint Security Assessment

Endpoint security protects individual computers from malware, unauthorized access, and network-based attacks. The assessment checked Microsoft Defender Antivirus, Windows Firewall, and disk encryption.

Assessment

Security Control	Status
Microsoft Defender Antivirus	Enabled
Real-Time Protection	Enabled
Current Threats	None
Windows Firewall	Enabled
BitLocker Encryption	Disabled

Evidence

7. Windows Security Antivirus Status

Virus & threat protection

Protection for your device against threats.

AVG Antivirus

AVG Antivirus is turned on.

Current threats

 No actions needed.

Protection settings

 No actions needed.

Protection updates

 No actions needed.

[Open app](#)

8. Firewall Status

🔒 Firewall & network protection

Who and what can access your networks.

🏢✅ Domain network

No action needed.

🏠✅ Private network

No action needed.

💻✅ Public network

No action needed.

Allow an app through firewall

9. [Screenshot] BitLocker Status

[Control Panel Home](#)

BitLocker Drive Encryption

Help protect your files and folders from unauthorized access by protecting your drives with BitLocker.

Operating system drive

C: BitLocker off



Turn on BitLocker

Fixed data drives

New Volume (D:) BitLocker off



Removable data drives - BitLocker To Go

Insert a removable USB flash drive to use BitLocker To Go.

Findings

Finding 3: BitLocker Disabled

Risk Level: Medium

BitLocker provides full-disk encryption that protects data if a device is lost or stolen. Since encryption is disabled, stored data may be accessible if unauthorized users gain physical access to the drive.

Recommendation: Enable BitLocker or Device Encryption if supported.

5.4 User Security Practices

User behavior plays an important role in cybersecurity. Unsafe downloads, weak security habits, and excessive permissions can increase security risks.

Assessment

Security Control	Status
Downloads Folder Review	Completed
Suspicious Files	Not Confirmed
Browser Extensions	Reviewed
Startup Applications	Reviewed
Account Type	Administrator

Downloads Assessment

Several executable files (.exe) were found in the Downloads folder, including software installers.

Example files:

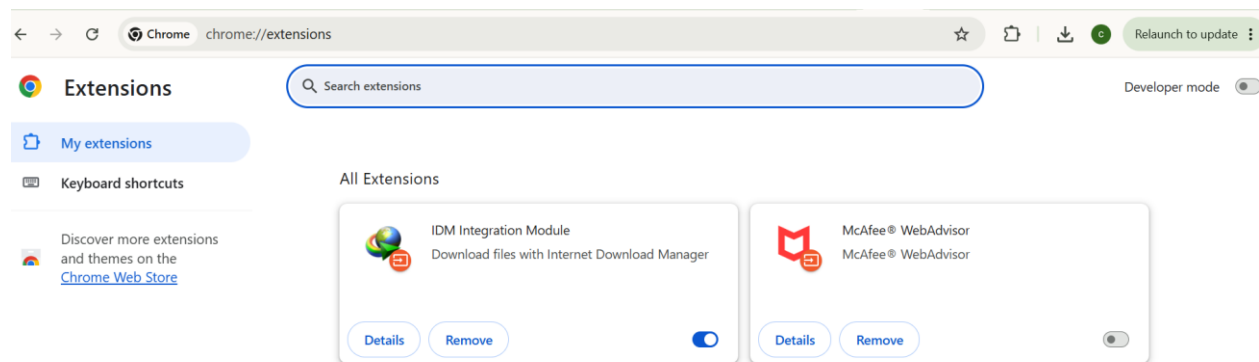
- PotPlayer Setup.exe
- Other executable files

No confirmed malicious files were identified.

10. Downloads Folder

Name	Date modified	Type	Size
Earlier this week (1)			
[REDACTED].exe	7/7/2026 10:27 AM	Application	63,916 KB
Last week (1)			
DOC-20260501-MA0101 (1).docx	7/3/2026 5:12 PM	Microsoft Word Doc...	32 KB
Last month (8)			
[REDACTED]	6/26/2026 12:50 PM	Microsoft Word Doc...	2,160 KB
[REDACTED].exe	6/25/2026 12:52 PM	Application	196,550 KB
[REDACTED].exe	6/23/2026 2:11 PM	Application	1,285 KB
[REDACTED]	6/22/2026 12:12 PM	Microsoft PowerPoint...	295 KB
[REDACTED]	6/22/2026 12:01 PM	Microsoft PowerPoint...	295 KB
(Show all 8)			
Earlier this year (63)			
[REDACTED].exe	5/31/2026 8:23 PM	Application	54,307 KB
[REDACTED].exe	5/31/2026 6:47 PM	Application	2,081 KB
Introduction-to-STATISTICAL-THEORY-BSc-Part...	5/23/2026 3:19 PM	Adobe Acrobat Docu...	4,202 KB
tx_alg2_01_05.pdf	5/23/2026 3:07 PM	Adobe Acrobat Docu...	977 KB
Functional English Final term notes.pdf	5/23/2026 12:54 AM	Adobe Acrobat Docu...	835 KB
(Show all 63)			
A long time ago (111)			
[REDACTED].exe	12/18/2025 7:02 PM	Application	1,877 KB
[REDACTED].exe	12/18/2025 7:01 PM	Application	1,877 KB
[REDACTED].exe	12/18/2025 7:01 PM	Application	1,877 KB
[REDACTED].exe	12/18/2025 7:00 PM	Application	1,877 KB
[REDACTED].4.exe	12/18/2025 6:58 PM	Application	1,878 KB

11. Browser extensions



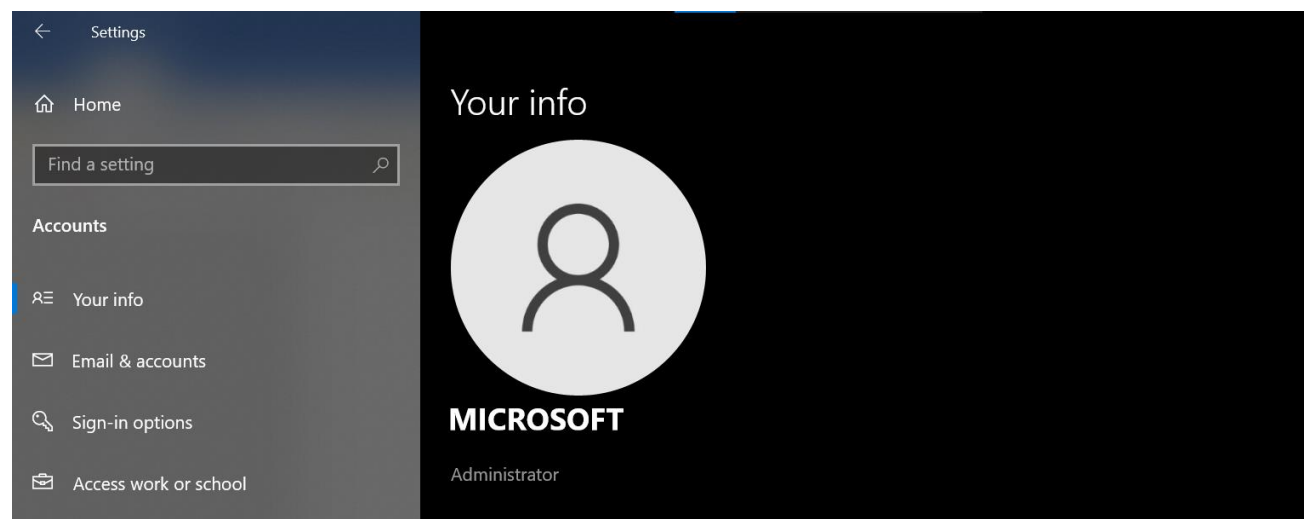
Startup Applications

Check	Result
Total Startup Applications	17
Disabled Applications	3
Unknown Applications	None

12. Startup Apps

Name	Publisher	Status	Startup impact
 Intel® Graphics Command ...	INTEL CORP	Disabled	None
 Phone Link	Microsoft Corporation	Disabled	None
 AVG AvLaunch component	Gen Digital Inc.	Enabled	Not measured
 Canva	Canva Pty Ltd	Enabled	Not measured
 epicpen	Tank Studios Limited	Disabled	Not measured
 Microsoft 365 Copilot	Microsoft Corporation	Enabled	Not measured
 Microsoft Edge	Microsoft Corporation	Enabled	Not measured
 Realtek HD Audio Universal ...	Realtek Semiconductor	Enabled	Not measured
 Send to OneNote Tool	Microsoft Corporation	Enabled	Not measured
 Update	Discord Inc.	Enabled	Not measured
 updatechecker.exe		Enabled	Not measured
 VMware Tray Process	VMware, Inc.	Enabled	Not measured
 Waves MaxxAudio Service A...	Waves Audio Ltd.	Enabled	Not measured
 WhatsApp	WhatsApp Inc.	Enabled	Not measured
 Microsoft OneDrive	Microsoft Corporation	Enabled	Low
 Windows Security notificatio...	Microsoft Corporation	Enabled	Low
 Internet Download Manager...	Tonec Inc.	Enabled	Medium

13. Account type



Findings

Finding 4: Administrator Account Used Daily

Risk Level: **Medium**

Using an administrator account for daily activities increases risk because malicious

applications may obtain elevated permission

Recommendation: Use a standard user account for regular activities and use administrator privileges only when required.

5.5 User Account Control (UAC)

User Account Control is a Windows security feature that prevents unauthorized system changes by requesting permission before administrative actions.

Assessment

Security Control	Status
UAC Setting	Never Notify

Evidence

14. UAC Settings

Choose when to be notified about changes to your computer

User Account Control helps prevent potentially harmful programs from making changes to your computer.

[Tell me more about User Account Control settings](#)

Always notify



Never notify

Never notify me when:

- Apps try to install software or make changes to my computer
- I make changes to Windows settings



Not recommended.

Findings

Finding 5: UAC Disabled

Risk Level: High

UAC is configured as "Never Notify", meaning Windows will not warn the user when applications attempt to make administrative changes. This reduces protection

against unauthorized modifications.

Recommendation: Change UAC to the default setting: "Notify me only when apps try to make changes to my computer."

6. Risk Summary Table

Finding	Risk Level
No Windows Password	High
UAC Disabled	High
MFA Disabled	Medium
BitLocker Disabled	Medium
Administrator Account Usage	Medium
Windows Updates Enabled	Low
Antivirus Enabled	Low
Firewall Enabled	Low
No Unknown Startup Programs	Low

7. Recommendations

Based on the assessment, the following improvements are recommended:

1. Create a strong Windows password.
2. Enable MFA for online accounts.
3. Enable BitLocker disk encryption.
4. Change UAC from "Never Notify" to the default security level.
5. Use a standard user account for daily activities.
6. Continue maintaining Windows updates.
7. Keep Microsoft Defender and Firewall enabled.
8. Regularly review installed software and startup applications.
9. Download software only from trusted sources.

8. Conclusion

The Windows system demonstrates a good basic security posture because important protection mechanisms such as Windows Update, Microsoft Defender Antivirus, and Firewall are enabled.

However, several security weaknesses were identified, including the absence of a Windows password, disabled Multi-Factor Authentication, disabled BitLocker encryption, and disabled User Account Control.

By implementing the recommended security improvements, the overall security level of the system can be significantly improved and the risk of unauthorized access or data compromise can be reduced.